Structural Non-Isolability in Knowledge-Based Robot Fault Diagnosis: Tie-Break Intervals and Grouped Validation

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Abstract: On the public voraus-AD robot benchmark we evaluate a diagnosis layer that scores the 14 failure modes of a hand-written FMEA knowledge base against per-signal deviations from normal operation. In 514 of 755 anomalous cycles two or three modes share the top score, always inside one of four groups whose declared signatures are identical or contain one another. Over the 12 tie-break orders that matter, top-1 accuracy ranges from 0.119 to 0.200; a prevalence prior that reads no evidence scores 0.187 and is better on top-3, top-5, mean reciprocal rank and group accuracy under every order. We therefore report the non-isolable group and search statistics of the existing signals for one that separates it. For the load-torque group the sign of the deviation separates the members (held-out accuracy 0.840 against 0.562); the other groups are inconclusive, undetermined or untestable. The search must be validated on unseen fault instantiations: under a random split over cycles it adopts a repair for the collision group that leave-one-setting-out validation adopts on only 4 of 9 folds, below the majority-class rate. A flat classifier over the same features is more accurate than the whole pipeline once more than a few labels are available.

Keywords: Fault isolation and root cause analysis, Structural analysis methods, Diagnosability analysis, Knowledge-based diagnosis, Anomaly detection, Robotics.

1. Introduction

An anomaly detector on robot telemetry can report that a cycle is abnormal, but not why. A common way to add a cause is a line-specific FMEA knowledge base: each failure mode declares the joints it affects and the signal families in which it shows, a retrieval function scores the modes against the per-signal deviations of the cycle, and the maintenance technician receives a ranked list of candidate causes.

This paper measures what such a list can decide on the public voraus-AD benchmark (Brockmann et al., 2024). The declarations of the knowledge base form a fault signature matrix in the sense of Blanke et al. (2016). In this matrix nine of the fourteen failure modes share their signature with another mode: six have identical columns, and in a group of three one column contains the other two. The scoring function is continuous, but it does not separate these modes. It gives them exactly the same score: in 514 of 755 anomalies two or three modes share the highest score, and the reported answer is then decided by the order in which the modes are listed (Fig. 1).

Two consequences do not depend on this knowledge base. The first concerns the reference for judging accuracy. Our earlier evaluation of this layer compared top-1 accuracy with uniform chance. When fault frequencies are skewed, the appropriate reference is a prevalence prior, which ranks the modes by their frequency in the training data and gives this ranking for every cycle. On voraus-AD the prior is better than the retrieval on every measure that does not depend on the tie-break order (Sec. 5).

The second concerns validation. The `setting` field of voraus-AD identifies a physical fault instantiation, for example one added weight or one placement of a collision object, and each instantiation is repeated over several cycles. A random split over cycles puts cycles of the same instantiation into both the training and the test data. We show that this changes which learned repair is adopted, not only the reported accuracy (Sec. 7). Leakage of this kind is known in machine-learning-based science (Kapoor and Narayanan, 2023) and in fault-diagnosis benchmarks (Hendriks et al., 2022); here it enters through the inner validation that selects the repair. We test both points on voraus-AD only.

Contributions. (i) The knowledge base analysed as a fault signature matrix, with non-isolable groups reported instead of broken by a tie-break rule. (ii) Two references for a ranked candidate list on this benchmark: a prevalence prior and the exact range of top-1 accuracy over tie-break orders. (iii) A search over named statistics of the existing signals for one that separates a group, reported as a certificate with held-out evidence, controls and a cost class. (iv) Evidence that grouping the validation by fault instantiation changes

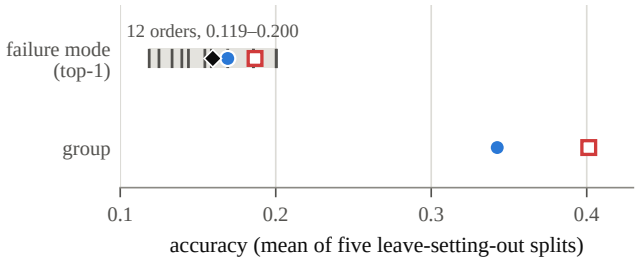


Figure 1: Top-1 and group accuracy of the retrieval, means over the splits of Table 1. Upper row: band and ticks, all 12 relative orders of the tied modes; circle, declared order; diamond, mean over orders; square, prevalence prior. Lower row: the group (circle) is the same under every order; always naming the most frequent group (square) is more accurate.

which repair is adopted. (v) A flat classifier that outperforms our pipeline, with the label budget from which it does.

2. Fault signatures and isolability

Structural diagnosability analysis determines, before any data are seen, which faults a diagnosis system can separate, using the incidence structure of tests and fault modes (Blanke et al., 2016; Krysander et al., 2008). The standard object is the fault signature matrix, whose rows are tests or residuals and whose columns are fault modes. Under the exoneration assumption two modes are non-isolable exactly when their columns coincide. Without it isolability is directional: a mode whose column is contained in another mode's column cannot be isolated from that mode (Krysander and Frisk, 2008; Cordier et al., 2004). The analysis has been applied to public process benchmarks (Düstegör et al., 2006), extended to graded distinguishability under noise (Jung et al., 2015), and is still developed (Travé-Massuyès and Wotawa, 2024).

When the signatures do not separate the faults, both FDI signature matching and consistency-based diagnosis in the DX tradition (Reiter, 1987; de Kleer and Williams, 1987) return a set of candidates, and the two coincide up to the exoneration assumption (Cordier et al., 2004; Travé-Massuyès, 2014). A ranked list that breaks ties hides this set; we report it as the non-isolable group of the top-ranked mode.

The usual remedy for a non-isolable group is sensor placement: find an additional measurement that makes the group isolable, at the lowest cost (Krysander and Frisk, 2008; Frisk et al., 2009). Our setting has no analytical model of the cell, and the rows of the signature matrix are declarations in an FMEA knowledge base (affected joints, signal families, cycle phase), not residuals of an observer. What we can search are statistics of the existing signals that the deployed representation does not use. This is test selection with data-driven evaluation, close to the hybrid scheme of Jung et al. (2018). Only its hardware cost class corresponds to sensor placement, and its evidence is held-out accuracy on unseen fault instantiations, not a structural guarantee.

3. System and data

Benchmark. voraus-AD (Brockmann et al., 2024) contains machine data from a six-axis robot on a pick-and-place task, with 12 anomaly types physically induced. We use the 100 Hz release: 2122 cycles, of which 755 are anomalous and 1367 normal. The official training data are the 948 normal cycles recorded under `setting` = 72; the remaining 419 normal cycles are held out. The anomalous cycles fall under 72 `setting` values, each identifying one physical fault instantiation repeated for 4 to 20 cycles (median 10). Three categories have a single instantiation.

Evidence record. Diagnosis is evaluated on all 755 labelled anomalous cycles, that is, assuming perfect detection; no number below depends on a detector. (A normalizing-flow detector of the family of Brockmann et al. (2024) reaches AUROC 0.854 on the official split; it would select the cycles in deployment.) Each of 26 channels (motor torque, torque sensor, joint position and joint velocity on six joints, robot and system current) is resampled to 500 points over task progress and standardised against the 948 normal training cycles, $z = (x - \mu)/\max(\sigma, 0.05\,\mathrm{range}\,\mu)$. The record of a cycle keeps the five channels with the largest peak $|z|$, the dominant joint (largest peak torque deviation in N m) and the action phase at that peak.

Knowledge base and retrieval. The knowledge base is an FMEA encoded as a graph. Liao et al. (2025) automate the construction of such graphs for robots; ours was written by hand by the authors from the voraus-AD anomaly descriptions and has not been reviewed independently. It declares 14 failure modes: 12 map one to one to the benchmark's anomaly types, and backlash and over-temperature have no instances. Each mode declares its joint scope, its signal families, a cycle-phase hint and a recommended check. A mode m is scored against a record by $s = J + F + 0.2E + 0.2P + 0.15C$, where J is 1 if the dominant joint lies in the scope of m (0.5 for scope 'any'), F is the Jaccard overlap of signal families, E is the mean over the record's channels of an IDF weight times $\log(1+|z|)$, counting only channels in the families and joints of m, P is 1 if the record's phase agrees with the phase hint of m, and C is the fraction of the families of m that are present. The scoring form and weights were settled during development on these 755 cycles. Choosing the three weights again on each training half of the protocol below gives a different choice on every split and no improvement on the test halves (MRR 0.362 against 0.366, top-5 0.600 against 0.651).

Protocol. Unless stated otherwise, results are means over five leave-setting-out splits: each `setting` is assigned wholly to the training half or to the test half,

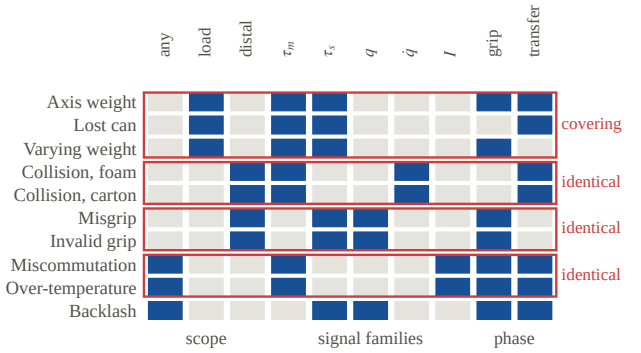


Figure 2: Knowledge-base columns of the nine modes in non-isolable groups and of backlash, which has the gripping pair's families but not their scope. Families: motor torque τ_m, torque sensor τ_s, joint position q, joint velocity $\dot{q}$, current I; a phase hint 'any' fills both phase cells.

stratified by category, so a learned component is always tested on instantiations it has not seen. Intervals are t-intervals over the five splits ($t_{0.975,4} = 2.776$). The splits re-partition the same 72 instantiations, so the intervals describe variation between splits on these data, not uncertainty about unseen instantiations. Comparisons between methods are made per split.

4. Non-isolable groups

We take the declared joint scope and signal families of a failure mode as its signature. Fig. 2 shows the signatures, with the phase hints, of the nine modes that share their signature with another mode. They form four groups: three load-torque modes (axis weight, lost can, varying weight), two distal collisions (foam, carton), two gripping faults (misgrip, invalid grip) and a commutation/over-temperature pair. Every term of the score except phase agreement depends only on the signature and the record. The score penalises extra families and a wider scope, which has the effect of the exoneration assumption for these rows, but the phase term rewards agreement without penalising disagreement. In the collision, gripping and commutation groups the phase hints are also equal, so the columns are identical and the members always receive the same score. In the load-torque group the hints differ (axis weight: any phase; lost can: transfer; varying weight: grip), so the column of axis weight contains the other two. Neither payload mode can then be isolated from axis weight, and axis weight is never isolated from the group.

In 514 of the 755 anomalies (68.1%) the top score is shared by two or three modes, and every tie lies inside one group. 498 ties are in the load-torque group: axis weight with lost can in 238 cycles, axis weight with varying weight in 199, and all three in 61 cycles whose phase has no hint class. The other 16 are commutation-pair ties. The count is 514 for every phase weight from 0 to 3; the weight changes only which members tie. These numbers describe this knowledge base, not the robot: declaring the phase of axis weight as transfer instead of any would leave 315 ties. What holds for any declarations is that no tie crosses a group. The collision and gripping groups never reach rank one, because they are declared on the wrist joints J4–J6, while the dominant joint, chosen by absolute torque deviation, is J1, J2 or J3 in every cycle.

We route each cycle by the margin between its top score and the best score outside its top group. Below $\tau = 0.15$, chosen on these cycles, it is *group-undetermined* (a reject option in the spirit of Chow (1970)); otherwise it is *decisive* if its top group has one member and *human-resolvable* if it has several. The routes take 122, 134 and 499 cycles, with group accuracy 0.180, 0.187 and 0.421. Of the rejected cycles 82% are in the wrong group (80% with τ chosen on training halves), against 66% of all cycles.

Table 1: Retrieval and references on the five test halves. Group: accuracy of the predicted mode's group; for the prior, the most frequent group of the training half. Balanced: mean per-category recall, with tied modes sharing the credit; a constant scores $1/K$, with K the 10 or 11 categories in a test half. In parentheses: top-1 with tied modes sharing the credit. Two-stage: Sec. 6; flat classifier: Sec. 8.

	top-1	top-5	group	balanced
uniform chance	0.071	0.357	0.150	0.083
prevalence prior	0.187	0.705	0.401	0.095
retrieval	0.169 (0.160)	0.651	0.342	0.105
two-stage	0.309 (0.302)	—	0.342	0.208
flat classifier	0.525	—	0.574	0.397

5. What the reported accuracy does not show

5.1. The prevalence prior

Against uniform chance, 0.071 over 14 modes, the deployed retrieval has a top-1 lift of 2.37×, the figure we first reported. Uniform chance is the wrong reference when categories differ in frequency: category 1 contributes 156 cycles and category 8 contributes 10. The prevalence prior ranks the modes by their frequency in the training half and gives this ranking for every cycle; it reads no evidence.

Table 1 compares the two. On top-3, top-5, MRR and group accuracy the prior is ahead under every tie-break order of Sec. 5.2; for top-3, MRR and group accuracy it is ahead on all five splits. On top-1 the order decides. Under the declared order the retrieval trails by 0.018 and is behind on four of five splits; under another order it is ahead. Breaking ties by training-half frequency, which uses only the information the prior uses, gives 0.200, level with the prior. Group accuracy does not favour the retrieval either: always naming the most frequent group of the training half, which is load torque

on every split, scores 0.401 against 0.342.

Balanced recall, the mean of the per-category recalls, is the measure a constant predictor cannot raise: a constant scores $1/K$, which averages 0.095. The retrieval's balanced recall also depends on the tie-break, so we let tied modes share the credit. The retrieval then reaches 0.105, above the constant on every split but by only 0.003 to 0.016, and one order puts it below. It carries per-cycle information, but little: its expected recall is zero on six of the twelve categories. Comparisons between methods below use balanced recall in this form.

5.2. The tie-break interval

The scores do not determine which member of a tied set is reported; our implementation reports the member declared first. Only five modes ever tie, always within their group, so only $3! \cdot 2! = 12$ relative orders of the modes can change the result. Fig. 1 shows top-1 accuracy under all of them. It ranges from 0.119 to 0.200, a width of 0.082. The declared order gives 0.169, and the mean over the orders, 0.160, equals the top-1 accuracy obtained when tied modes share the credit (McSherry and Najork, 2008). Two of the twelve orders reach the prior's 0.187.

This spread is larger than the one due to the choice of split: over the orders the five-split mean has standard deviation 0.027 and range 0.082, while the declared-order value has standard deviation 0.015 and range 0.040 over the five splits. More splits would not reduce the first, because one order applies to all splits. Both ranges exceed the 0.018 by which the retrieval trails the prior. An oracle that always picks the correct member of the tied set reaches 0.281. This bounds rules that choose inside the tied set, but not rules that choose inside the whole group, because the phase term has already removed the true mode from 46 tied sets.

The group of the top-ranked mode does not depend on the order: group accuracy is 0.342 under every tie-break rule, because no tie crosses a group. The group is fixed, but it is not accurate. As a set it contains the true mode in 34.2% of cycles with 2.34 modes on average, while the same number of modes taken from the top of the prevalence list contains it in 38.5%. Only on balanced coverage is the group ahead (0.243 against 0.222). The knowledge base fixes how coarse the answer is, and the evidence does little to choose the group.

6. Test selection over representation dimensions

For each non-isolable group we ask which statistic of the existing signals, if the representation carried it, would separate the members.

Dimensions. The deployed representation keeps the peak magnitude $\max|z|$ of each of the 26 record channels (*mag*). The candidate dimensions are further statistics of the same z: *sign* (mean of z), *step* (last third of the cycle minus first third), *payload* (carrying minus empty actions), *shape* (standard deviation of z) and *unread* (peak, signed and payload statistics of two channel families that are recorded but not read by the attribution). Each has a cost class: *free* if computed from the record, *pipeline* if it needs channels the attribution does not read, *hardware* if it needs a measurement the cell does not have. None of our dimensions is of hardware class.

Search and certificate. Every learned component is a standardised ℓ_2 logistic regression ($C = 1$). Starting from mag, a greedy search adds the dimension with the largest gain in grouped inner cross-validation accuracy on the training half ($k = \min(4, \text{settings per class})$ folds), as long as the gain is at least 0.005 times its cost factor (1 for free, 6 for pipeline). The selected set is adopted on a split if its inner accuracy exceeds the training-half majority rate by 0.05 and that of mag alone by 0.02; a group is adopted if this holds on at least three of five splits. If a member has fewer than two instantiations in the training half, no search can run; we then evaluate the set of all free dimensions and do not adopt. The test half is only reported: the certificate gives the dimensions, their cost class, the held-out accuracy within the group and three controls (Table 2). Varying the cost factor, the minimum gain and both margins does not change which group is adopted (repository).

Adopted. Within the load-torque group the certificate reaches 0.840 on held-out instantiations, against 0.562 for the deployed features, a majority rate of 0.504 and 0.431 with permuted labels, and it is adopted on all five splits. The dimension selected most often is sign (4 of 5 splits; step on 3, unread and shape on 2 each), which mag does not contain. Because unread enters on two splits, the certificate's worst-case cost class is pipeline, but the free set {sign, step} reaches 0.869 against 0.872 for the best set with unread channels, so no additional channel is needed. Most remaining errors are between the two payload modes: on cycles of these two modes the resolver is right in 0.80 of cases, against 0.93 and 0.98 for the pairs with axis weight. These figures are computed on cycles whose true mode is in the group, so they state how well the repair works once a cycle reaches the group (Sec. 8.2).

Not adopted. Carton collision has three instantiations, so each training half contains one and no grouped search can run. The set of all free dimensions reaches 0.776 against a majority rate of 0.718. Each split trains on one carton instantiation and tests on the other two, and three of the five splits test on the same two, so the splits are not independent replications. Sec. 7 compares the protocols on this group with folds on which the search can run.

Undetermined. Each gripping mode has one instantiation, so no grouped split can test anything.

Untestable. voraus-AD contains no over-temperature fault, so this verdict is assigned, not searched. The knowledge base names a separator for this pair, current against torque for the same motion,

Table 2: Repair certificates: accuracy within the group on held-out cycles of its members. Grouped: leave-setting-out, the protocol we report. Ungrouped: the same code under a random split over cycles, including its inner validation. Deployed: the same classifier on the mag features alone. Majority: the training-half majority rate. Permuted: the selected dimensions refitted on training labels shuffled over cycles. Deployed, Majority and Permuted are grouped. †One carton instantiation in each training half, so no grouped search can run; the value is for the set of all free dimensions.

Group	Ungrouped	Grouped	Deployed	Majority	Permuted	Verdict
load torque (3 modes)	0.933	0.840	0.562	0.504	0.431	**adopted** on 5/5 splits, both protocols
distal collision (2 modes)	0.830	0.776†	0.709	0.718	0.608	**not adopted**: search cannot run; adopted ungrouped
gripping (2 modes)	one instantiation per member					undetermined
commutation/temp (2 modes)	no over-temperature instance in voraus-AD					untestable

which would be of pipeline cost.

End to end, the two-stage system (retrieval, then the resolver for cycles whose top group is load torque) raises top-1 from 0.169 ± 0.018 to 0.302 ± 0.047 with dimensions searched on each split. Fixing the dimensions that most splits chose, {sign, step}, gives 0.309 ± 0.031, balanced recall 0.208 (from 0.105) and 0.869 within the group. That choice used all five splits; made from the other four splits each time, it gives 0.301 and 0.846, so the choice is worth less than 0.01. The gain over the retrieval is positive on every split, from 0.089 to 0.151. It comes from the resolver, not from the routing: sending every cycle to the load-torque resolver gives 0.349 and balanced recall 0.237, more than the two-stage system on all five splits.

7. The validation protocol changes a deployment decision

Each `setting` value is one physical fault instantiation. A random split over cycles therefore places cycles of the same instantiation in both halves, and a model can learn to recognise the instantiation instead of the fault. The methods react to the protocol as their use of labels predicts. The flat classifier of Sec. 8 falls from 0.800 under a random split to 0.525 under leave-setting-out. The retrieval, which uses no labels, does not change (0.161 and 0.162 over 20 splits of each kind).

For the load-torque group the certificate is adopted under both protocols (Table 2). For the collision group the protocols disagree. Under a random split the inner validation, whose folds share instantiations, reaches 0.85 to 0.96, and the repair is adopted on all five splits, although on held-out cycles it beats the majority class of the same split by only 3 of 47 cycles (0.830 against 0.766). Under the 50/50 leave-setting-out split no grouped search can run (Sec. 6), so we compare the protocols with nine folds and the same search and rule. With folds stratified over cycles, the repair is adopted on 9 of 9 folds. With one setting held out per fold, it is adopted on 4 of 9 folds, and its pooled held-out accuracy is 0.67 to 0.75 over ten inner seeds, below the majority rate of 0.766 for every seed.

A permuted-label control that shuffles labels over cycles does not detect the difference, because the shuffled labels are still learned from cycles of the same instantiations. Permuting labels over whole settings does: under the random split, between 5 and 10 of the 84 possible labellings reach the observed collision accuracy on each split ($p = 0.06$ to 0.12), while none of 300 labellings reaches the load-torque accuracy ($p < 0.004$). Any learned comparison on this benchmark should therefore group by setting in the inner validation as well as in the outer split.

8. Controls, and what would settle the open verdicts

8.1. The flat classifier

A flat 12-way logistic regression over all dimensions reaches 0.525 top-1, 0.397 balanced recall and 0.574 group accuracy under the same protocol, against 0.309, 0.208 and 0.342 for our system. We tested whether the knowledge base at least saves labels. Over 20 paired leave-setting-out splits, the classifier is trained on k labelled cycles and the two-stage system on k labelled cycles per group; only the load-torque group receives cycles, so only its labels are used. Both are scored on the twelve-way task. At $k = 3$ the two-stage system, which is then close to the knowledge base alone, has the higher balanced recall on 17 of 20 splits. From $k = 8$ the classifier has the higher top-1 (0.225 against 0.208, 15 of 20 splits), from $k = 12$ also the higher balanced recall (17 of 20), and at $k = 320$ it reaches 0.550 against 0.304. If the two-stage system receives the same random labels as the classifier, it is behind at $k = 8$ on all 20 splits. Where more than a few labels can be collected, the flat classifier is the better choice; the advantage of the knowledge base is limited to the case with almost no labels.

8.2. Where the ranking's output goes

The load-torque group is the true answer for 41.1% of cycles but is ranked first for 66.0%. Of the cycles it

receives, 58% are not load-torque faults. The resolver is right on 0.88 of the true members it receives, so the route as a whole is right on 0.37, against 0.16 for the retrieval alone on the same cycles. The collision and gripping groups are the true answer for 12.5% and 3.0% of cycles and are ranked first for none (Sec. 4), so for them there is no route to measure; their certificates state whether a repair would work if the group were reached. The flat classifier recognises foam collision in 0.506 of cases, a category on which the retrieval and the prior both score zero, so the features contain evidence that the attribution and the retrieval do not use.

8.3. What would settle the open verdicts

Gripping has one instantiation per mode, so no grouped protocol can evaluate it. For collision the limit is our split: a 50/50 grouped split leaves $\lfloor N/2 \rfloor$ instantiations per mode for training, the grouped inner search needs two, and carton has $N = 3$. With one setting held out at a time the search runs and is still not adopted (Sec. 7). Sweeps over N on injected faults and on a simulated cell are in the repository; in both the deployed features already separate the pair, so they do not show how many instantiations a real non-isolable pair needs.

9. Conclusion

On voraus-AD the ranked list of an FMEA-based diagnosis layer is limited by its knowledge base before any data are scored: 68.1% of anomalies tie at rank one, almost all in the load-torque group. A single top-1 figure then depends on the tie-break order more than on the method, and a prevalence prior is better on every measure that does not depend on the order.

We therefore report the non-isolable group and search for a statistic that separates it, with held-out evidence, controls and a cost class. One of the three groups with benchmark instances is separated on unseen instantiations, by the sign of the deviation at no extra cost. This search must be validated by instantiation: a random split adopts a collision repair that leave-setting-out validation does not, and a permuted-label control over cycles does not notice.

Whether a candidate list can decide can be checked before any data are seen, but a list that decides is not therefore more accurate. Adding the knowledge base's own declarations of torque direction and step to the score removes 513 of the 514 ties but lowers balanced recall on four of five splits (0.105 to 0.100) and top-1 on all five. A change of representation should be adopted on evidence from unseen instantiations, not because it makes the list decidable.

Code and data. Evidence records, certificates, all analyses and the producing scripts are archived at `https://doi.org/10.5281/zenodo.22089173`.

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